ModelIt!

Teacher's Guide

Quantum Computing: Why 0 and 1 Aren't Enough

Modeling the Physics and Engineering of Computation Beyond Classical Limits

Quick Reference

Grade Level	Grade 10L3
NGSS Standard	HS-PS4-3, HS-PS2-5
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

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1 NGSS Standards & CAST Alignment

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Performance Expectation: HS-PS4-3, HS-PS2-5

Evaluate the claims, evidence, and reasoning behind the idea that electromagnetic radiation can be described either by a wave model or a particle model. Plan and conduct an investigation to provide evidence that an electric current can produce a magnetic field and that a changing magnetic field can produce an electric current.

Three-Dimensional Learning

Science Practice	**Evaluating Claims and Evidence** — Students evaluate claims about quantum computing capabilities by modeling the physical constraints that determine whether quantum advantage is achievable for specific problems.
Disciplinary Core Idea	**PS4.A Wave Properties / PS2.B Types of Interactions** — Students model how quantum mechanical properties (superposition, entanglement) enable fundamentally different information processing, and how electromagnetic interactions cause decoherence.
Crosscutting Concept	**Cause and Effect** — Students trace causal chains from environmental noise through decoherence to error accumulation to computational failure, identifying the mechanisms that limit quantum computing performance.

CAST Testing Alignment

MULTIPLE CHOICE:

A quantum computer's coherence time is 100 microseconds and each gate operation takes 50 nanoseconds. If a useful computation requires 5,000 gates, what does your model predict about the computation's success without error correction?

CONSTRUCTED RESPONSE:

Using your model, explain why doubling the number of qubits does not simply double a quantum computer's power. Address the roles of coherence time, error rate, and error correction overhead in your explanation.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

Classical computers have transformed civilization using a simple language: 0 and 1. Every photo, video, app, and AI model is built from billions of binary switches. But some problems are so complex that even a computer with every atom in the universe as a transistor could not solve them before the heat death of the cosmos. Quantum computing offers a fundamentally different approach — but building it requires engineering at the edge of physical possibility.

Classical vs. Quantum Information

A classical bit is a switch: 0 or 1, on or off. A qubit exploits quantum superposition to be in a combination of 0 and 1 simultaneously, described by probability amplitudes. Two qubits can represent four states at once, three can represent eight, and N qubits can represent 2^N states simultaneously. For 300 qubits, that is more states than atoms in the observable universe. This exponential scaling is the source of quantum computing's potential power.

The Decoherence Problem

Quantum states are extraordinarily fragile. Any interaction with the environment — a single stray photon, a vibration from a passing truck, thermal energy from a fraction of a degree of warmth — can cause decoherence, collapsing the delicate superposition into an ordinary classical state. This is why quantum computers are cooled to 15 millikelvin (colder than outer space), suspended on vibration-isolating platforms, and shielded from electromagnetic radiation. Even with all these precautions, coherence times are measured in microseconds.

The Error Correction Challenge

With per-gate error rates around 0.1-1%, a quantum circuit with 1,000 gates has a near-certain probability of containing errors. Quantum error correction encodes one logical qubit across many physical qubits, allowing errors to be detected and corrected. But the overhead is staggering: current schemes require roughly 1,000 physical qubits per logical qubit. A useful quantum computer might need millions of physical qubits to produce the thousands of error-corrected logical qubits needed for practical algorithms.

The Race to Quantum Advantage

In 2019, Google claimed quantum supremacy with its 53-qubit Sycamore processor, completing a specific calculation in 200 seconds that would take a classical supercomputer 10,000 years. Critics argued the problem was artificial. True quantum advantage — solving a practical, economically valuable problem faster than any classical approach — remains the holy grail. The most promising applications are drug discovery (simulating molecular interactions), cryptography (breaking current encryption), optimization (logistics, finance), and materials science (designing new materials at the atomic level).

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**

Students identify nine key components of a quantum computing system: qubit coherence time, entanglement fidelity, error rate, gate operation speed, cooling temperature, quantum volume, decoherence rate, classical overhead, and computational advantage.

E **Establish Relationships**

Students map relationships showing that cooling temperature drives decoherence rate, which determines coherence time, which limits the number of gate operations possible — while error rate determines how much of that limited time must be spent on error correction rather than useful computation.

V **Visualize & Model**

Students build a computational model showing the nine-component quantum system with the central tension between computation speed and decoherence.

E **Evaluate Outcomes**

Students run current technology, near-perfect qubits, and thermal disruption scenarios to identify the parameter thresholds where quantum advantage becomes achievable.

R **Revise & Extend**

Students add qubit connectivity, readout fidelity, or crosstalk interference to model more realistic quantum processor behavior.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"Your phone has billions of classical bits. This machine has 1,000 qubits and costs 50 million dollars. Yet for certain problems, this machine wins. Why?"*

Watch For

Show an image of a quantum computer's golden chandelier hardware next to a smartphone. The contrast hooks students: why is this bizarre machine more powerful?

Slide 2: Learning Objectives

3 min

Say: *"Today you are modeling computation at the boundary of physics — where the rules of quantum mechanics create possibilities that classical physics cannot match, but also create engineering nightmares."*

Watch For

Have students read objectives. Pre-teach 'superposition' and 'decoherence' using the coin analogy: a spinning coin is both heads and tails until it lands.

Slide 3: Big Question

3 min

Say: *"A classical computer with 300 bits can be in one of 2^300 states. A quantum computer with 300 qubits can be in all 2^300 states simultaneously. That number has more digits than atoms in the universe."*

Watch For

Quick-write: Students estimate how many states 10 classical bits vs. 10 qubits can represent. Share out. Then scale to 50, 100, 300 — watch the exponential growth click.

Slide 4: LEVER Framework

2 min

Say: *"We are building a model of the most delicate machine ever engineered — a computer that operates at temperatures colder than deep space and breaks if a truck drives by on the street outside."*

Watch For

Preview LEVER steps. Emphasize that this model captures the central tension: quantum power versus quantum fragility.

Slide 5: Activity 1: Components

8 min

Say: *"Nine parameters. Two you can control through engineering. Seven that the physics determines. The question is whether engineering can push the physics far enough."*

Watch For

Guide students through components. Discuss why Cooling Temperature and Classical Overhead are external — they represent the engineering resources committed to supporting quantum operations.

Slide 6: Activity 2: Connections

8 min

Say: *"Here is the cruel paradox: errors require correction, correction requires more qubits, more qubits generate more errors. How do you break that cycle?"*

Watch For

Help students map the decoherence-error-correction cascade. Identify the error correction overhead paradox. Use red for destructive pathways.

Slide 7: Activity 3: Simulation

10 min

Say: *"Can today's quantum computers actually outperform a classical supercomputer? Let us find out — and discover exactly what needs to improve."*

Watch For

Students predict Computational Advantage for each scenario. Run simulations. Compare how dramatically near-perfect qubits change the picture.

Slide 8: Discoveries

5 min

Say: *"Quantum computing is not about making computers faster — it is about making a fundamentally different kind of computation possible. But only if engineering can keep the quantum states alive long enough."*

Watch For

Lead discussion connecting model findings to real quantum milestones. Google's quantum supremacy claim, IBM's quantum roadmap, the path to useful quantum advantage.

Slide 9: STEM Challenge

12 min

Say: *"You have 1,000 physical qubits. How many become computation qubits, how many become error correction, and what can you actually solve? Your strategy determines whether this is a 50-million-dollar paperweight or a revolutionary machine."*

Watch For

Groups design error correction allocation strategies. Present with justification from model data. Class evaluates which strategy maximizes real computational power.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Cooling Temperature, Classical Overhead
- **INTERNAL:** Qubit Coherence Time, Entanglement Fidelity, Error Rate, Gate Operation Speed, Quantum Volume, Decoherence Rate, Computational Advantage

Relationships

- **Cooling Temperature to Decoherence Rate:** NEGATIVE (-) — Lower cooling temperature means fewer thermal photons and phonons interacting with qubits, reducing the rate at which environmental noise destroys quantum states. Even tiny temperature increases dramatically accelerate decoherence.
- **Decoherence Rate to Qubit Coherence Time:** NEGATIVE (-) — Higher decoherence rate means quantum states collapse faster, directly reducing the time window available for quantum operations. Coherence time is inversely related to decoherence rate.
- **Error Rate to Classical Overhead:** POSITIVE (+) — Higher error rates require more physical qubits dedicated to error correction codes and more classical processing for syndrome measurement and correction — increasing the classical overhead and reducing the qubits available for actual computation.

Key Discoveries

- Quantum computing power is not limited by processing speed but by the race between computation and decoherence — every calculation must finish before the quantum state collapses
- Error rates compound exponentially across quantum circuits, meaning a 0.5% error per gate becomes near-certain failure after a few hundred gates without error correction
- Error correction itself requires massive overhead — correcting one logical qubit may need 1,000+ physical qubits, creating a resource paradox where most qubits are spent on error management rather than computation
- Quantum advantage is problem-specific, not universal — quantum computers will not replace classical computers but will solve specific problems (cryptography, molecular simulation, optimization) that are fundamentally intractable classically

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Quantum computers are just faster classical computers"	Quantum computers are fundamentally different, not just faster. They do not execute classical operations more quickly — they exploit superposition and entanglement to explore solution spaces in ways impossible for classical architecture. For most everyday tasks (email, web browsing, video streaming), quantum computers offer no advantage. Their power emerges only for specific problem types where quantum algorithms exist. A quantum computer will never replace your laptop — it will solve problems your laptop cannot even attempt.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Qubits are just bits that can be 0 and 1 at the same time"	This is a common oversimplification that misses the crucial point. A qubit in superposition is described by a complex probability amplitude — a mathematical object with both magnitude and phase. Quantum advantage comes not just from superposition but from interference, where probability amplitudes add or cancel. Quantum algorithms are carefully designed so that correct answers constructively interfere (amplitudes add) and wrong answers destructively interfere (amplitudes cancel). Without this interference engineering, superposition alone is useless.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Quantum computers will break all encryption tomorrow"	Current quantum computers have fewer than 1,500 noisy physical qubits. Breaking RSA-2048 encryption with Shor's algorithm would require approximately 4,000 error-corrected logical qubits, which translates to roughly 20 million physical qubits at current error rates. Even optimistic timelines place this capability a decade or more in the future. Meanwhile, post-quantum cryptography algorithms that are resistant to quantum attack are already being standardized and deployed. The threat is real but not imminent.